

Sarthak Vajpayee

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SUMMARY

Accomplished Data Scientist with 5+ years in NLP, and deep learning, driving innovation in data-driven environments. Proven in developing AI/ML solutions and extracting valuable insights from unstructured data, with expertise in prompt engineering and GPU computing. Skilled in leveraging cutting-edge technologies with strong communication skills to deliver high-impact projects. Seeking to join a forward-thinking organization with a strong willingness to relocate to drive business growth and customer satisfaction.

PROFESSIONAL EXPERIENCE

University of Texas at Dallas – Richardson, Texas, USA

Jan 2024 - May 2024

Data Science Research Assistant

- Boosted research efficiency to 10x by building a RAG that leverages LLMs, LangChain, Pinecone Vector Database, and text-embeddings to analyze 200+ prompt engineering studies, extracted insights that inform evidence-based decisions.
- Surpassed top existing sentiment classification research by 20% through fine-tuning transformer models including T5, BERT, Mistral, and LLaMA with PEFT techniques such as QLoRA, using Hugging Face, PyTorch, BitsAndBytes, and Python.

AppSteer – Dallas, Texas, USA

May 2023 - Dec 2023

Data Scientist (Full-Time CPT)

- Catalyzed \$200K in revenue by engineering an Agent-based Q&A ChatBot that reduced error rates by 50% through prompt engineering using LLMs, LangChain, Python, PySpark, and PostgreSQL.
- Accelerated time-to-market from 4 days to 2 minutes by reducing the application development and deployment timelines through automation using FastAPI, Pydantic, Docker, LangChain, HuggingFace, PyTorch, and OpenAI on Azure.
- Enhanced deployment rate by 20% through orchestration of over 50 cloud ETL deployments, enhancing CI/CD workflows with Git version control, Jenkins, Kubernetes and Prometheus.

Ernst & Young – Bengaluru, India

Dec 2021 - Jul 2022

Data Scientist

- Drove \$1.1M in annual cost savings by enhancing sales forecast models through data processing with Azure Databricks and implementing feature engineering techniques with Azure ML, resulting in an 8% improvement in demand forecast accuracy.
- Established new benchmarks in international market analytics by developing a second-generation predictive system that combined ARIMA and XGBoost in PySpark for advanced data modeling, achieving a MAPE of 10% within 15 days.
- Achieved a 15% increase in forecast accuracy by identifying critical demand trends through time-series analysis and A/B testing, leveraging Tableau dashboards for enhanced data visualization.

Scaler – Hyderabad, India

Sep 2018 - Nov 2021

Data Science Engineer

- Boosted resolution efficiency by 25% by architecting a robust LMS and online coding platform using FastAPI, Kubernetes, AWS, and Docker, integrating an NLP-based ticketing system with Python.
- Led the initiative to develop an end-to-end plagiarism detection tool, improving detection accuracy by 30% using Doc2Vec, FastText, and BERT with Python and TensorFlow.

EDUCATION

The University of Texas at Dallas, Texas

Aug 2022 - May 2024

Master of Science - Business Analytics (Specialization in Data Science)

Dr. A.P.J. Abdul Kalam Technical University, India

Aug 2014 - Jun 2018

Bachelor of Technology - Electronics & Communication Engineering

PROJECT EXPERIENCE

Global Top 5% in Google Quest Q&A Challenge

- Secured a Top 5% position on the Kaggle leaderboard for the Google Quest Q&A Labeling Challenge by creating an ensemble of transformer models (BERT, ALBERT, RoBERTa, XLNet) to predict StackExchange question-answer ratings.

CI/CD for Credit Risk Prediction

- Developed and deployed an ETL pipeline and Streamlit app for Credit Risk Analysis using Python, Decision Trees, PostgreSQL, Jenkins, Docker, enabling 20% faster credit decisions and improved efficiency through CI/CD.

COVID-19 Diagnostics through Transfer Learning

- Led a groundbreaking research project using transfer learning to enhance a RESNET-50-based COVID-19 diagnostic tool, achieving 0.96 AUC and positioning it as a formidable alternative to RT-PCR tests in accuracy.

SKILLS

Programming: Python, R, SQL, NoSQL, MongoDB, JavaScript, C, MATLAB, Bash, SAS, Linux/Unix, VBA.

Libraries: Scikit-Learn, Spacy, NumPy, Pandas, Keras, PyTorch, Flask, PySpark, FAISS, ChromaDB, Matplotlib, TensorFlow.

Software & Tools: Git, Docker, Kubernetes, Amazon Web Services (AWS), Azure, Google Cloud Platform (GCP), Hadoop, Hive, Snowflake, Apache Spark, Scala, Tableau, Excel, Jira, Kafka, Confluence, SageMaker, Airflow, MLflow, Power BI, CUDA, KPI.

Courses: Advanced Statistics, Econometrics and Time Series Analysis, Big Data, Data Warehousing, Cloud Computing, NLP.